

Memory Is Expressivity: A Hankel-Rank Barrier for State-Space Models on Graph Random Walks

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Abstract—Walk-based graph learners feed random walks, decorated with random node features, into a sequence model; recognising a cycle of length k then reduces to deciding whether the token at time t equals the token at time $t - k$. We show that this primitive imposes a sharp memory requirement on linear time-invariant (LTI) state-space models (SSMs): with real state dimension $S < k$, the Hankel matrix of any lag- k probe has rank at most S while its target has k unit singular values, so the probe error has $\mathcal{H}_\infty$ norm at least one and mean-squared error at least $1/k$; the bound is tight at $S = k$. On circular skip-link (CSL) graphs, where 1-WL message passing is at chance, we compute exactly the non-backtracking return horizon $W^*(s)$ that separates each class; an exact-count estimator with window S recognises every one of the ten classes at the first tested $S \geq W^*(s)$. A direct fit attains the bound at $S = k$, but end-to-end trained diagonal SSMs need a median of $4.0\times$ the Hankel minimum for reliable token-level lag detection, input-dependent (selective) dynamics do not reduce this, and on CSL the best trained model reaches 60.4% against 98.3% for the oracle.

Index Terms—State-space models, graph neural networks, Hankel operators, expressivity, random walks.

I. INTRODUCTION

MESSAGE-PASSING graph neural networks (GNNs) are bounded by the 1-dimensional Weisfeiler–Leman (1-WL) test [1], [2] and cannot, for instance, separate the regular circular skip-link (CSL) graphs [3], [4]. Walk-based learners circumvent this by reading random walks as sequences [5], [6], and random node features make walk revisits observable [7], [8]. With structured state-space models (SSMs) [9]–[11] now used as the sequence backbone for graphs [6], [12], [13], a basic question arises: *how much state does a walk-based SSM need to see a cycle?*

CRAWI [5] fixes an explicit window s and provably detects cycles up to length s ; an SSM has no window, only a state. We answer the question with classical realisation theory [14]. Our contributions are: (i) an exact impossibility result and a tight quantitative bound, via Hankel-rank and Nehari/AAK arguments [15]–[17], showing that an LTI SSM needs real state $S \geq k$ to probe lag- k revisits; (ii) an exact, graph-level prediction of the state each CSL class needs, computed from the non-backtracking return profile [18]; (iii) experiments separating the *information* barrier from the *optimisation* overhead of gradient-trained LTI and selective SSMs. Our results

complement the copying lower bounds of [19] by giving a signal-processing characterisation specific to graph walks.

II. PROBLEM SETTING

Let $G = (V, E)$ be a graph and $v_0, v_1, \dots, v_T$ a non-backtracking (NB) walk ($v_{t+1} \neq v_{t-1}$). Each walk receives fresh features $r_v \sim \mathcal{N}(0, I_d)$, i.i.d. over v , and the model reads $x_t = r_{v_t}$. A walk-based SSM with real state $s_t \in \mathbb{R}^S$ is

$$s_t = A_t s_{t-1} + B_t x_t, \quad \hat{y}_t = g(s_t, x_t), \quad (1)$$

with arbitrary readout g . The model is *LTI* if $A_t \equiv A$, $B_t \equiv B$, and *selective* if (A_t, B_t) depend on x_t as in Mamba [11]. The *lag- k revisit* label is $y_t = \mathbf{1}[v_t = v_{t-k}]$; on NB walks, $y_t = 1$ exactly when the last k steps close an NB cycle, so revisit statistics are cycle statistics. Because features are continuous, $y_t = \mathbf{1}[x_t = x_{t-k}]$ almost surely.

III. THEORY

Proposition 1 (Exact detection). *If $S < k$, then for every LTI pair (A, B) and every readout g there exist two token sequences with the same (s_t, x_t) but $y_t = 1$ and $y'_t = 0$.*

Proof: By Cayley–Hamilton, $A^S = \sum_{i=0}^{S-1} c_i A^i$, hence $A^k = \sum_{i=0}^{S-1} c_i A^{i+k-S}$, where the exponents lie in $\{k-S, \dots, k-1\} \subseteq \{1, \dots, k-1\}$. Fix any sequence with $x_{t-k} = x_t$ and any $e \neq 0$. Set $x'_{t-k} = x_{t-k} + e$ and $x'_{t-j} = x_{t-j} - c_{j-k+S} e$ for $j \in \{k-S, \dots, k-1\}$, leaving all other tokens (including x_t) unchanged. Then $s_t - s'_t = \sum_{j \geq 1} A^j B(x_{t-j} - x'_{t-j}) = -A^k B e + \sum_i c_i A^{i+k-S} B e = 0$, while $y'_t = 0$. ■

Proposition 1 is a worst-case statement. The next result quantifies how badly any linear read-out of the state must fail on *typical* inputs. A lag- k probe $z_t = c^\top s_t + b^\top x_t$ estimates $w^\top x_{t-k}$ ($\|w\| = 1$). Its error $z_t - w^\top x_{t-k} = \sum_{m \geq 0} e_m x_{t-m}$ has impulse response $e_0 = c^\top B + b^\top$ (which can be cancelled by b) and $e_m = c^\top A^m B - \delta_{mk} w^\top$ for $m \geq 1$, with transfer function $E(z) = \sum_{m \geq 1} e_m z^{-m}$.

Theorem 1 (Hankel barrier). *If $S < k$, every LTI probe satisfies*

$$\sup_\omega \|E(e^{j\omega})\| \geq 1, \quad (2)$$

$$\sum_{m=1}^{2k-1} \min(m, 2k-m) \|e_m\|^2 \geq 1, \quad (3)$$

and hence, for white unit-variance tokens, $\mathbb{E}[(z_t - w^\top x_{t-k})^2] \geq 1/k$. Conversely, for $S = k$ and every $\varepsilon > 0$ there is an LTI probe with mean-squared error below ε .

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Code and data: <https://github.com/The-MathKing/ITE>.

Proof: Let $\mathcal{H}_k(f)$ be the $k \times kd$ block-Hankel section with (i, j) block f_{i+j+1} , $0 \leq i, j < k$. For the model, $h_{i+j+1} = (c^\top A^i)(A^{j+1}B)$, so $\text{rank } \mathcal{H}_k(h) \leq S$. The target $\delta_{m,k}w^\top$ yields a block anti-diagonal matrix with k singular values equal to $\|w\| = 1$. By Eckart–Young, $\|\mathcal{H}_k(e)\|_2 \geq \sigma_{S+1}(\mathcal{H}_k(\delta w^\top)) = 1$. A finite section is dominated by the Hankel operator norm, which by Nehari’s theorem [15] is at most $\|E\|_\infty$; this gives (2). Bounding the spectral norm by the Frobenius norm, where e_m occurs $\min(m, 2k - m)$ times, gives (3); and $\min(\cdot) \leq k$ gives the mean-squared bound. For tightness, take $A = \rho P$ with P the k -cycle permutation, $B = e_1 w^\top$, $c = \rho^{-k} e_1$, and b to cancel e_0 . Then $e_m = \rho^{m-k} w^\top$ for $m = 2k, 3k, \dots$ and 0 otherwise, so the error is $\rho^{2k}/(1 - \rho^{2k}) \rightarrow 0$ as $\rho \rightarrow 0$. ■

Remark 1. The tight construction needs gains of order ρ^{-k} , i.e. the near-optimal LTI probe is exponentially ill-conditioned in k . The bound also limits *precision*, not *information*: for $S < k$ a probe can still be partially correlated with the lag- k token ($\text{MSE} \geq 1/k < 1$), so statistics pooled over long walks can remain informative below the threshold. Selective SSMs break the factorisation $h_{i+j+1} = (c^\top A^i)(A^{j+1}B)$, so Theorem 1 does not apply to them; we test them empirically.

A. Graph-level consequence: NB return horizons

Let $\rho_G(\ell) = \Pr[v_\ell = v_0]$ for the stationary NB walk. We compute it exactly from the NB (Hashimoto) operator on directed edges. Theorem 1 implies that an LTI walk-SSM with state S has (approximate) linear access to revisits only at lags $\ell \leq S$. For a family $\{G_s\}$ we define the *distinguishability horizon*

$$W^*(s) = \min \left\{ W : \max_{\ell \leq W} |\rho_{G_s}(\ell) - \rho_{G_{s'}}(\ell)| > \tau \quad \forall s' \neq s \right\}, \quad (4)$$

with $\tau = 2 \cdot 10^{-3}$. For the ten CSL classes ($n = 41$, skip $s \in \{2, 3, 4, 5, 6, 9, 11, 12, 13, 16\}$), $W^*(s)$ ranges from 3 to 8 (Table I). The prediction is that class s becomes recognisable once $S \geq W^*(s)$.

IV. EXPERIMENTS

Models. We use one diagonal complex SSM layer with S4D-Lin initialisation [10]. It has $\lfloor S/2 \rfloor$ complex modes plus one real mode if S is odd, so its real state dimension is exactly S . The discretisation is $A = \exp(\Delta\Lambda)$ and $B_t = \Delta B$. The LTI variant learns Δ per mode; the *selective* variant sets $\Delta_t = \text{softplus}(W_\Delta x_t + b)$. The readout g sees $[\Re s_t, \Im s_t, x_t]$ and comparison features $(Qx_t - Ks_t)^{\odot 2}$, followed by a two-layer MLP. It is a pointwise map of (s_t, x_t) and is therefore covered by Proposition 1. Features are $d = 8$ dimensional; training uses AdamW with a cosine schedule. All runs are on CPU, and the code regenerates every figure and number in this letter.

Delay realisation. We first fit the impulse response of an S -state diagonal LTI system to the pure delay $\delta_{m,k}$ ($m \geq 1$) by gradient descent, keeping the best of four restarts (Fig. 1). Below the bound ($S < k$), the median relative ℓ_2 error is 0.81 (the minimum is 0.45). Above it, the median is 0.01. An error of 0.1 is first reached at a median of $1.0k$ states, so the Hankel bound is attained in practice when the delay is fitted directly.

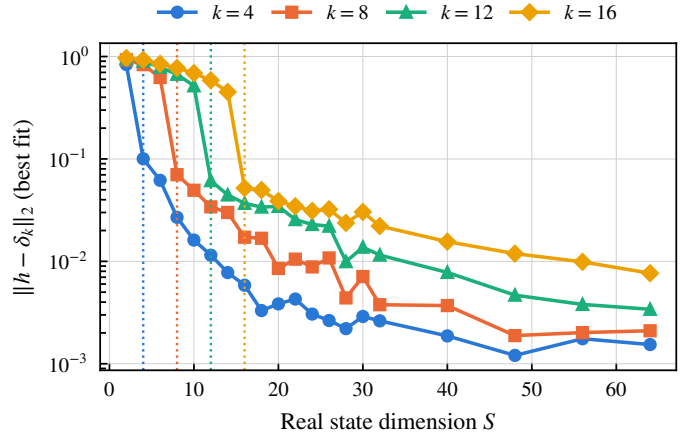


Fig. 1. Best gradient fit of an S -state diagonal LTI impulse response to the lag- k delay. Dotted lines mark the Hankel threshold $S = k$.

Token-level phase diagram. We train on NB walks of length 64 over 256 random 4-regular graphs ($n = 24$), built as unions of two Hamiltonian cycles, and test on 64 unseen graphs. Fig. 2 reports test AUROC for the lag- k revisit label. Below the bound ($S < k$), the mean AUROC is 0.61 (LTI) and 0.61 (selective). For $S \geq 4k$ it is 0.99 and 0.99. Where an AUROC of 0.95 is reached, it takes a median of $4.0\times$ (LTI) and $3.4\times$ (selective) the Hankel minimum $S = k$. This holds for all 9 lags up to $k = 16$, so the requirement grows linearly in k as Theorem 1 predicts, but with a constant roughly four times larger than in the direct fit of Fig. 1. The overhead is thus a property of end-to-end training through the readout rather than of realisability. Below the bound, AUROC stays above 0.5, which reflects the partial correlation allowed by the Remark. Selectivity changes AUROC by $+0.000$ on average (mean absolute difference 0.008), so input-dependent dynamics do not escape the memory requirement.

CSL classification. Each training example is one NB walk (length 128) on a CSL graph, and the class is predicted from mean-pooled features. At test time, each graph is judged from 16 walks of length 256. As a reference we use an *oracle* that applies multinomial logistic regression to the exact revisit frequencies at lags $1, \dots, S$. This is the best a window- S reader can do. Figs. 3 and 4 and Table I summarise three seeds. The oracle reaches 98.3% and exceeds 95% from $S = 8$. For all 10 classes, its onset is exactly the first tested $S \geq W^*(s)$ (the grid contains even S only), which confirms the horizon analysis. Trained walk-SSMs rise well above the 10% 1-WL level but remain far below the oracle: the best LTI model reaches $60.4 \pm 2.6\%$ at $S = 64$, and the best selective model reaches $55.2 \pm 2.8\%$ at $S = 64$. Only 3 (LTI) and 3 (selective) classes reach 80% accuracy at any $S \leq 64$. One of them ($s = 2$) does so below its horizon, which is possible because pooling 16×256 tokens amplifies the partial correlation left by the Remark. On graphs, therefore, the binding constraint in practice is not the Hankel capacity but the optimisation of the precise, ill-conditioned delays that Theorem 1 requires for every class-specific lag simultaneously.

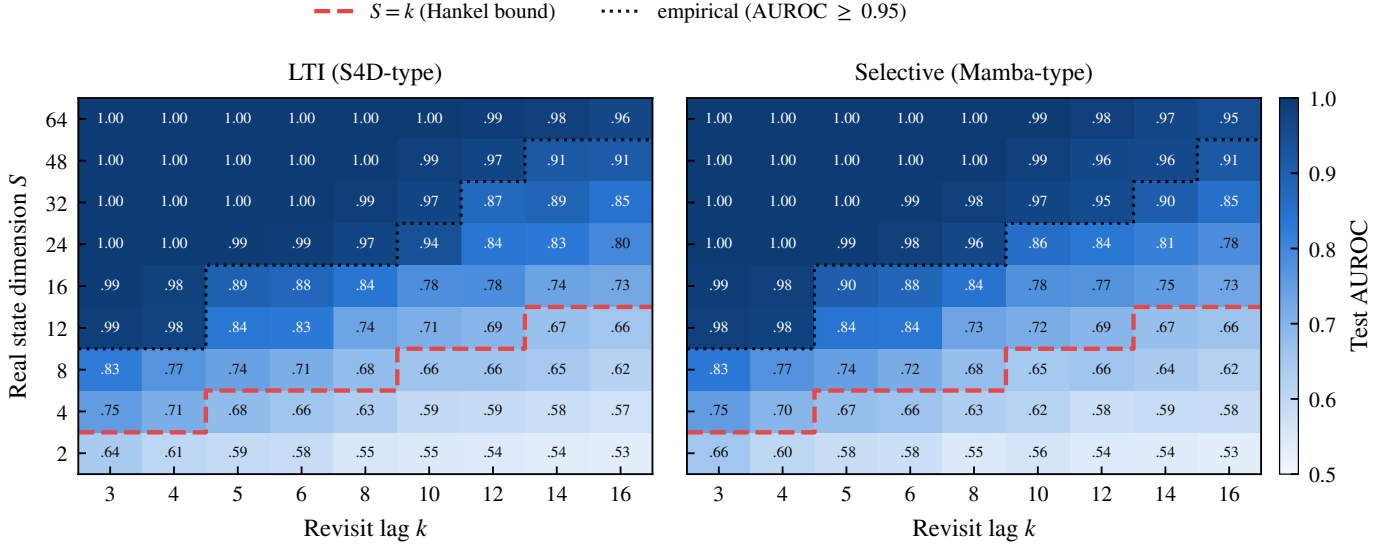


Fig. 2. Test AUROC for lag- k revisit detection on NB walks over random 4-regular graphs. Red dashed line: Hankel bound $S = k$ (Theorem 1). Dotted line: first S reaching an AUROC of 0.95.

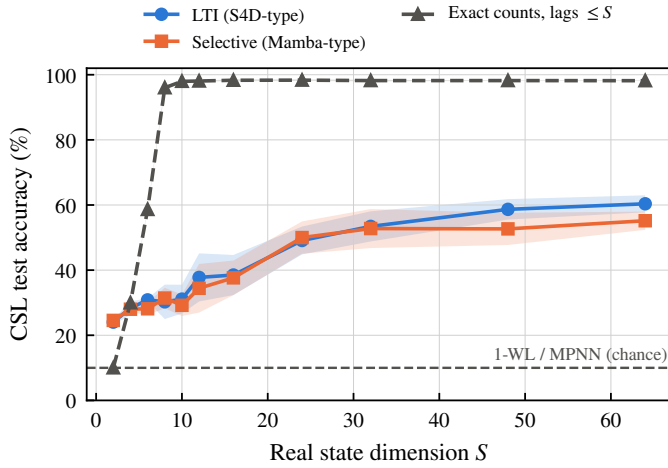


Fig. 3. CSL test accuracy against real state dimension (mean $\pm$ std over three seeds). The oracle uses exact revisit counts at lags $\leq S$.

TABLE I
CSL HORIZONS $W^*(s)$ AND OBSERVED PER-CLASS ONSETS S (“-”: NOT REACHED).

s	2	3	4	5	6	9	11	12	13	16
$W^*(s)$	3	4	5	6	7	8	7	8	5	7
Oracle	4	4	6	6	8	8	8	8	6	8
LTI	2	6	-	-	-	-	-	-	32	-
Selective	2	8	-	-	-	-	-	-	64	-

V. CONCLUSION

For walk-based graph learners with LTI state-space backbones, seeing a cycle of length k costs at least k real state dimensions. This is a Hankel-rank fact with a tight $\mathcal{H}_\infty$ and mean-squared form. Combined with exact non-backtracking return horizons, it predicts the state size at which each CSL class becomes separable, and an exact-count oracle

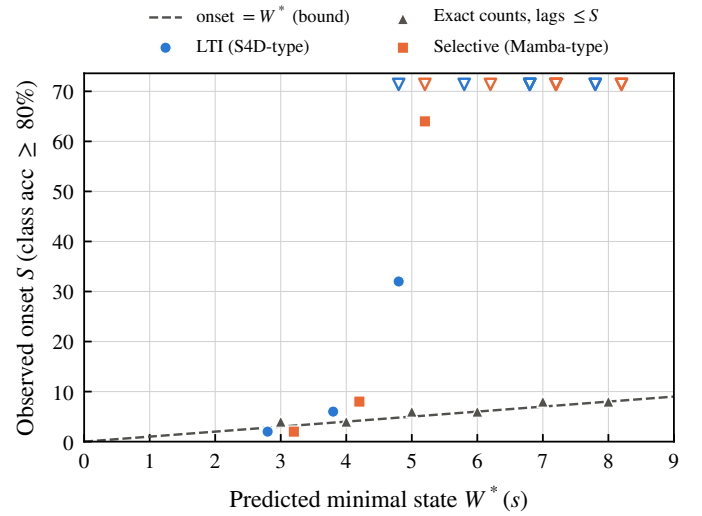


Fig. 4. Per-class onset (smallest S with class accuracy $\geq 80\%$) against the predicted horizon $W^*(s)$. Hollow triangles: classes never reaching 80% for $S \leq 64$.

confirms the prediction for all ten classes. Direct fits attain the bound, but end-to-end trained diagonal SSMs, with or without selectivity, need about four times the minimal state for token-level detection and stay far from the oracle on CSL. This points to the optimisation of precise delays, rather than capacity alone, as the practical bottleneck. Natural extensions are lower bounds for selective SSMs and better-conditioned delay parameterisations for graph SSMs.

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